$\partial C_\ell/\partial\omega_m$ at fixed θ_s: CosmoPower emulators (analytic) vs. CLASS (numerical)

Omega_m binning project

19 October 2026

1 What is compared

All derivatives are taken at the same fiducial point,

$$\omega_b = 0.02237, \quad \omega_{\rm cdm} = 0.12, \quad h = 0.674, \quad \tau = 0.054, \quad n_s = 0.965, \quad \ln(10^{10}A_s) = 3.044,$$

with one massive neutrino of 0.06 eV ($N_{\rm ur} = 2.0328$) and HMcode nonlinear lensing, i.e. the physics both emulators were trained on. $\omega_m = \omega_b + \omega_{\rm cdm} + \omega_\nu$ with ω_ν fixed, so at fixed ω_b a change of ω_m is a change of $\omega_{\rm cdm}$. All spectra are lensed TT, TE, EE in $\mu{\rm K}^2$, $2 \le \ell \le 2508$. The quantity is

$$\left.\frac{\partial C_\ell}{\partial\omega_m}\right|_{\theta_s,\,\omega_b,\,\tau,\,n_s,\,A_s}, \qquad 100\,\theta_s = 1.0419693546 \text{ (CLASS definition, } r_s/D_A \text{ at the visibility peak).} \tag{1}$$

Five versions are computed:

1. **CLASS, numerical** (benchmark). CLASS v3.4.1, five-point stencil

$$\frac{\partial C_\ell}{\partial\omega_m} \simeq \frac{C_\ell(-2\epsilon) - 8C_\ell(-\epsilon) + 8C_\ell(\epsilon) - C_\ell(2\epsilon)}{12\,\epsilon}, \qquad \epsilon = 1.2e-03, \tag{2}$$

   where $C_\ell(k\epsilon)$ is computed at $\omega_m + k\epsilon$ and at the h_k that solves $100\,\theta_s(\omega_b, \omega_m + k\epsilon, h_k) = 1.0419693546$ (root finding in h on CLASS's own θ_s, tolerance 10^{-12} in h). Precision: $\ell_{\rm max} = 3500$ (cut at 2508), `l_linstep`=5, `l_logstep`=1.04, `accurate_lensing`, `delta_l_max`=1000, refined transfer q-sampling and tightened perturbation tolerances (full list in `compute_derivs.py`).

2. **SPT emulator, analytic.** CosmoPower `SPT_high_accuracy` LCDM networks (Balkenhol et al. 2022, CAMB-trained). Exact derivative of the network (forward-mode chain rule through every layer, no step size), combined with the constraint slope:

$$\left.\frac{\partial C_\ell}{\partial\omega_m}\right|_{\theta_s} = \left.\frac{\partial C_\ell}{\partial\omega_{\rm cdm}}\right|_{h} + \left.\frac{\partial C_\ell}{\partial h}\right|_{\omega_{\rm cdm}} \left.\frac{dh}{d\omega_m}\right|_{\theta_s}, \qquad \left.\frac{dh}{d\omega_m}\right|_{\theta_s} = -\frac{\partial\theta_s/\partial\omega_m}{\partial\theta_s/\partial h} = -3.523928, \tag{3}$$

   the slope from CLASS (five-point derivatives of θ_s), i.e. $s \equiv d\ln h/d\ln\omega_m|_{\theta_s} = -0.74773$.

3. **SPT emulator, numerical.** Same stencil and same ($\omega_m + k\epsilon$, h_k) points as CLASS.

4. **lcdm v1 emulator (with_cp_lcdm), analytic** — the derivative the MCMC pipeline actually uses. CosmoPower lcdm v1 networks (Bolliet et al. 2023, CLASS-trained), evaluated through the pipeline's own `with_cp_lcdm/class/python/classy.py`, method `Class.dcl_domega_m()`: one exact Jacobian-vector product through the networks along $(d\omega_{\rm cdm}, dH_0) = (1,\, 100\,h\,s/\omega_m)$, with s from the pipeline's fit to CLASS, $s = -0.74844$ at this point (exact value -0.74773).

5. **lcdm v1 emulator, numerical.** Same stencil and points as CLASS, via the pipeline's `lensed_cl()`.

Error measure: model error relative to C_ℓ. In the linear-response model $C_\ell = C_\ell^{\Lambda{\rm CDM}} + (\partial C_\ell/\partial\omega_m)\,\delta\omega_m(\ell)$, an error $\Delta(\partial C_\ell/\partial\omega_m) = (\partial C_\ell/\partial\omega_m)_X - (\partial C_\ell/\partial\omega_m)_{\rm CLASS}$ in the template changes the model spectrum by

$$\Delta C_\ell^{\rm model} = \Delta(\partial C_\ell/\partial\omega_m)\,\delta\omega_m(\ell) \quad\Longrightarrow\quad \frac{\Delta C_\ell^{\rm model}}{C_\ell} = \underbrace{\frac{\Delta(\partial C_\ell/\partial\omega_m)}{C_\ell}}_{\text{plotted / tabulated}} \times\delta\omega_m(\ell). \tag{4}$$

Since $\delta\omega_m(\ell)$ is not known in advance, all relative errors are quoted *per unit* $\delta\omega_m$: $\Delta C_\ell^{\rm model}/(C_\ell\,\delta\omega_m)$. For TE the denominator is $\sqrt{C_\ell^{TT}C_\ell^{EE}}$, since C_ℓ^{TE} crosses zero. Example: a tabulated -4.0 (in units of 10^{-2}) means $\Delta C_\ell^{\rm model}/C_\ell = -0.04\,\delta\omega_m$, i.e. -0.04% of C_ℓ for $\delta\omega_m = 0.01$ and -0.005% for $\delta\omega_m = \sigma(\omega_m) = 0.0012$. The "CLASS numerical floor" is the larger of (i) the change of the CLASS derivative when the step is doubled and (ii) its change when the perturbation-tolerance settings are relaxed.

2 Is the CLASS derivative at fixed θ_s?

Yes. h is not held fixed: it is re-solved at every stencil point so that CLASS's $100\,\theta_s$ equals the fiducial value. Table 1 re-computes $100\,\theta_s$ independently at every (ω_m, h) pair used. The largest deviation is $4e-09$, i.e. 4×10^{-9} relative; propagated through the stencil this is a $\sim 3\times10^{-6}$ relative effect on the derivative, far below every other number in this note. Along the stencil h moves from 0.691 to 0.658 ($d\ln h/d\ln\omega_m \simeq -0.75$), as it must at fixed θ_s. The emulator derivatives use the same h_k (numerical) or the same slope (analytic), so all five are taken along the same fixed-θ_s direction.

Table 1: CLASS stencil points ($\omega_m = \omega_b + \omega_{\rm cdm} + \omega_\nu$, $\epsilon = 1.2e-03$) and CLASS's $100\,\theta_s$ re-evaluated at each of them. $k = \pm4$ are used only for the step-size check.

k	ω_m	h	$100\,\theta_s$	$100\,\theta_s - 100\,\theta_s^{\rm fid}$
-4	0.138 214	0.691 362 48	1.041 969 353 0	-1.6×10^{-9}
-2	0.140 614	0.682 567 77	1.041 969 354 3	-3.1×10^{-10}
-1	0.141 814	0.678 256 09	1.041 969 355 5	9.1×10^{-10}
+0	0.143 014	0.674 000 00	1.041 969 354 6	0.0
+1	0.144 214	0.669 798 18	1.041 969 350 6	-4.0×10^{-9}
+2	0.145 414	0.665 649 52	1.041 969 351 4	-3.2×10^{-9}
+4	0.147 814	0.657 506 15	1.041 969 351 1	-3.5×10^{-9}

3 Results

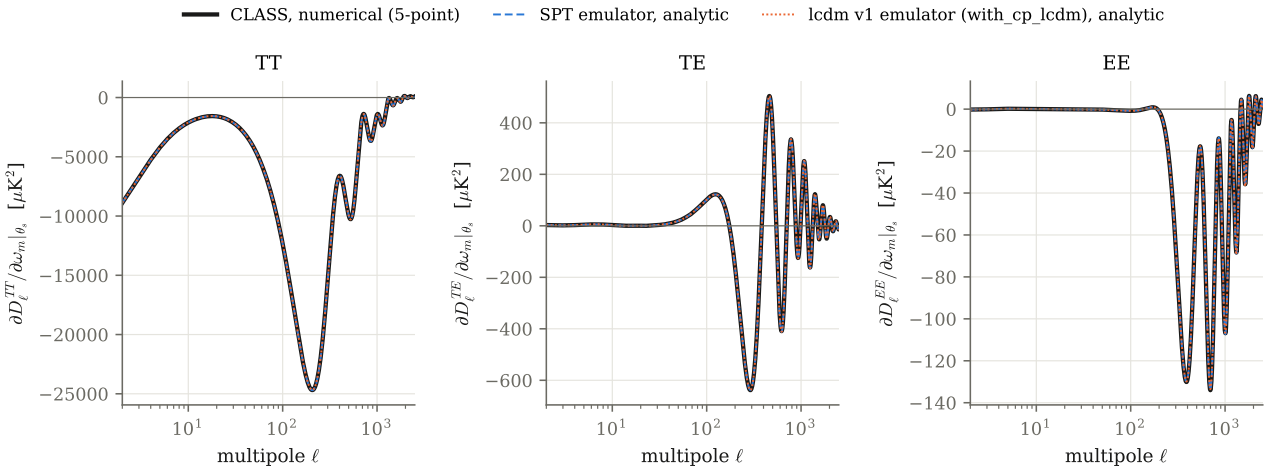


Figure 1: The derivative $\partial D_\ell/\partial\omega_m|_{\theta_s}$, $D_\ell = \ell(\ell+1)C_\ell/2\pi$, for TT, TE, EE: CLASS (numerical, black), SPT emulator (analytic, blue dashed) and lcdm v1 emulator of the with_cp_lcdm pipeline (analytic, orange dotted). The three are indistinguishable at this scale; differences in Fig. 2.

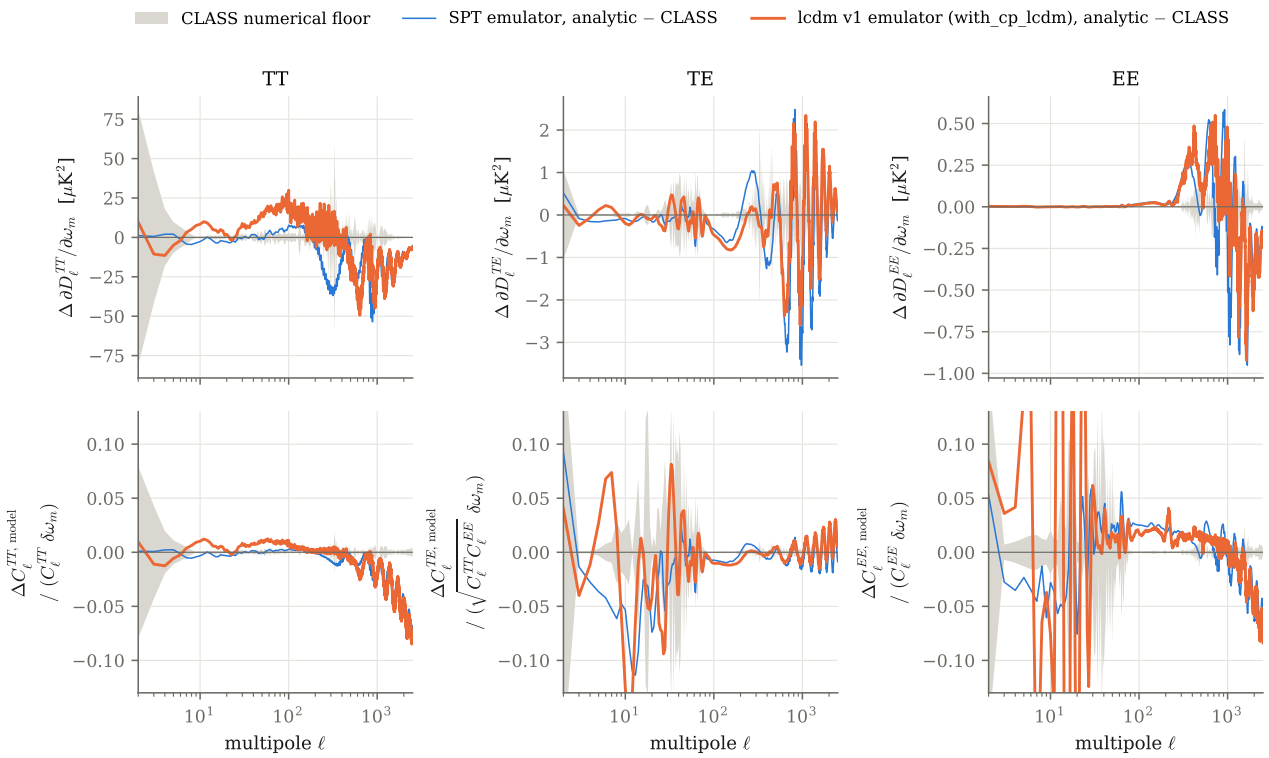


Figure 2: Error of the *analytic* emulator derivatives with respect to the CLASS benchmark. **Top:** absolute, $\Delta\,\partial D_\ell/\partial\omega_m$ in $\mu\mathrm{K}^2$ per unit ω_m. **Bottom:** the resulting model error relative to the spectrum, $\Delta C_\ell^{\mathrm{model}}/(C_\ell\,\delta\omega_m) = \Delta(\partial C_\ell/\partial\omega_m)/C_\ell$ (TE: $/\sqrt{C_\ell^{TT}C_\ell^{EE}}$); multiply by $\delta\omega_m$ to get $\Delta C_\ell^{\mathrm{model}}/C_\ell$. Orange (thick): lcdm v1 emulator, analytic, exactly as computed inside the with_cp_lcdm pipeline (`dcl_domega_m()`, fitted slope). Blue: SPT emulator, analytic. Grey band: CLASS numerical floor. Off scale in the bottom row: lcdm v1 TE at 2 multipoles with $\ell \le 12$ (max 0.16) and EE at 13 multipoles with $\ell \le 24$ (max 0.71), where C_ℓ^{EE} is at its minimum.

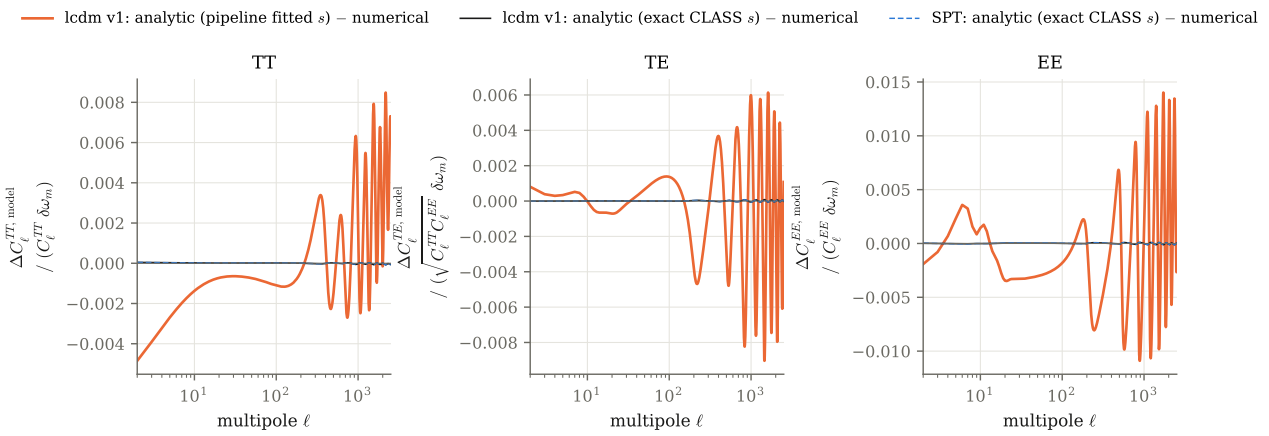


Figure 3: Analytic minus numerical derivative of the *same* emulator, as model error relative to the spectrum, $\Delta C_\ell^{\mathrm{model}}/(C_\ell\,\delta\omega_m)$. With the exact CLASS slope $s = -0.74773$ the analytic and numerical derivatives agree to $\lesssim 10^{-4}$ for both emulators (black and blue lines, on top of zero): the analytic Jacobians are exact. The orange line is the pipeline's analytic derivative, which uses the fitted slope $s = -0.74844$ (0.1% off): this alone shifts the model by up to $0.008\,\delta\omega_m$ (TT), $0.009\,\delta\omega_m$ (TE) and $0.014\,\delta\omega_m$ (EE) in units of C_ℓ.

Table 2: Noise-weighted size of each model error for $\delta\omega_m = \sigma(\omega_m) = 0.0012$ (scales linearly with $\delta\omega_m$): $\sqrt{\chi^2} = \left[\sum_\ell (\Delta\, \partial C_\ell/\partial\omega_m\, \delta\omega_m)^2/\mathrm{Var}(C_\ell)\right]^{1/2}$, cosmic variance only, $f_{\rm sky} = 0.57$, $30 \le \ell \le 2508$. The first row is the size of the template itself.

	TT	TE	EE
template itself (CLASS)	2.041	1.921	3.043
SPT analytic − CLASS	0.071	0.026	0.063
lcdm v1 analytic (pipeline) − CLASS	0.080	0.030	0.067
lcdm v1 analytic, exact s − CLASS	0.086	0.036	0.073
lcdm v1 analytic (pipeline) − numerical	0.007	0.010	0.012
CLASS numerical floor	0.005	0.006	0.009

Table 3: TT: model error $\Delta C_\ell^{\rm model}/(C_\ell^{TT}\,\delta\omega_m)$ in units of 10^{-2} (multiply by $\delta\omega_m$ for the fractional change of C_ℓ^{TT}). "lcdm v1" is the pipeline's analytic derivative (fitted s), "exact s" the same with the exact CLASS slope.

	analytic − CLASS			analytic − numerical		CLASS
ℓ	SPT	lcdm v1	lcdm v1, exact s	SPT	lcdm v1	floor
2	0.11	0.95	1.44	0.004	−0.485	7.94
10	−0.32	1.02	1.16	0.001	−0.137	0.09
30	−0.07	0.84	0.90	0.001	−0.064	0.25
100	0.28	1.11	1.22	0.001	−0.111	0.12
200	−0.03	0.09	0.13	0.000	−0.042	0.04
500	−0.39	−0.71	−0.56	0.001	−0.152	0.22
800	−0.90	−0.43	−0.35	0.001	−0.079	0.09
1000	−1.39	−2.00	−2.21	−0.002	0.207	0.09
1500	−4.02	−3.68	−4.28	−0.005	0.598	0.19
2000	−4.10	−4.27	−4.26	0.000	−0.013	0.15
2500	−6.90	−7.87	−8.59	−0.006	0.710	0.42

Table 4: TE: model error $\Delta C_\ell^{\rm model}/(\sqrt{C_\ell^{TT} C_\ell^{EE}}\,\delta\omega_m)$ in units of 10^{-2}. Columns as in Table 3.

	analytic − CLASS			analytic − numerical		CLASS
ℓ	SPT	lcdm v1	lcdm v1, exact s	SPT	lcdm v1	floor
2	9.25	4.01	3.93	−0.001	0.081	18.70
10	−5.32	−12.57	−12.56	0.000	−0.003	2.25
30	−3.05	0.24	0.26	0.000	−0.016	2.66
100	−0.73	−1.03	−1.16	−0.001	0.137	0.02
200	0.01	−0.79	−0.40	0.003	−0.385	0.12
500	0.26	0.45	0.76	0.003	−0.307	0.01
800	1.00	0.86	1.46	0.005	−0.596	0.17
1000	−1.27	−0.47	−1.07	−0.005	0.596	0.11
1500	−0.03	−0.24	0.34	0.005	−0.572	0.19
2000	1.39	2.27	2.32	0.000	−0.044	0.03
2500	−0.81	0.62	0.58	0.000	0.046	0.06

Table 5: EE: model error $\Delta C_\ell^{\rm model}/(C_\ell^{EE}\,\delta\omega_m)$ in units of 10^{-2}. Columns as in Table 3.

	analytic − CLASS			analytic − numerical		CLASS
ℓ	SPT	lcdm v1	lcdm v1, exact s	SPT	lcdm v1	floor
2	5.27	8.39	8.58	0.002	−0.192	16.60
10	−2.81	−7.59	−7.74	−0.001	0.144	2.03
30	2.41	6.18	6.52	0.003	−0.329	4.68
100	2.51	1.82	2.00	0.002	−0.181	0.09
200	1.97	1.72	1.69	0.000	0.035	0.18
500	−0.55	1.44	0.81	−0.006	0.630	0.28
800	−1.44	−0.40	−1.29	−0.008	0.887	0.43
1000	−1.12	0.03	−0.09	−0.001	0.114	0.64
1500	1.17	−0.73	0.11	0.007	−0.828	0.81
2000	−6.37	−5.67	−6.82	−0.010	1.140	0.26
2500	−5.26	−7.82	−7.56	0.002	−0.257	0.33

4 Summary

- The CLASS benchmark is a fixed-θ_s derivative to 4×10^{-9} in $100\,\theta_s$, with a numerical floor of $\sqrt{\chi^2}\leq 0.009$ (Table 2).
- Both emulators reproduce it to $\sqrt{\chi^2}\approx 0.03$–$0.09$ per 1σ shift, against a template of ≈ 2–3, i.e. to 1.5–4%. The model error grows with ℓ: $\Delta C_\ell^{TT,\rm model}/C_\ell^{TT}\approx -0.04\,\delta\omega_m$ at $\ell=1500$–2000 and $-(0.07$–$0.08)\,\delta\omega_m$ at $\ell=2500$, i.e. below 10^{-3} of C_ℓ at $\ell\geq 30$ for any $|\delta\omega_m|\lesssim 0.01$, for both emulators (similar shape, so likely shared training/settings differences with respect to this CLASS run).
- For the lcdm v1 (pipeline) derivative, the fitted slope s accounts for a visible part of the difference (Fig. 3); here it happens to partly cancel the emulator error ($\sqrt{\chi^2}$ 0.080/0.030/0.067 with the fitted slope vs 0.086/0.036/0.073 with the exact one).

A Values of the derivatives

Table 6: TT: D_ℓ and $\partial D_\ell/\partial\omega_m|_{\theta_s}$ from the five methods (μK^2).

ℓ	D_ℓ^{TT} [μK^2]	CLASS	SPT num.	SPT anal.	lcdm v1 num.	lcdm v1 anal.
		$\partial D_\ell^{TT}/\partial\omega_m\|_{\theta_s}$ [μK^2]				
2	1.022×10^3	-8.890×10^3	-8.889×10^3	-8.889×10^3	-8.875×10^3	-8.880×10^3
10	8.195×10^2	-2.112×10^3	-2.115×10^3	-2.115×10^3	-2.103×10^3	-2.104×10^3
30	1.057×10^3	-2.132×10^3	-2.133×10^3	-2.133×10^3	-2.123×10^3	-2.123×10^3
100	2.703×10^3	-1.234×10^4	-1.233×10^4	-1.233×10^4	-1.231×10^4	-1.231×10^4
200	5.607×10^3	-2.464×10^4	-2.464×10^4	-2.464×10^4	-2.464×10^4	-2.464×10^4
500	2.448×10^3	-9.947×10^3	-9.957×10^3	-9.957×10^3	-9.961×10^3	-9.964×10^3
800	2.526×10^3	-2.893×10^3	-2.916×10^3	-2.916×10^3	-2.902×10^3	-2.904×10^3
1000	1.062×10^3	-1.437×10^3	-1.451×10^3	-1.451×10^3	-1.460×10^3	-1.458×10^3
1500	6.880×10^2	-6.203×10^2	-6.479×10^2	-6.479×10^2	-6.497×10^2	-6.456×10^2
2000	2.312×10^2	8.479×10^1	7.530×10^1	7.530×10^1	7.494×10^1	7.491×10^1
2500	7.864×10^1	8.144×10^1	7.602×10^1	7.601×10^1	7.469×10^1	7.525×10^1

Table 7: TE: D_ℓ and $\partial D_\ell/\partial\omega_m|_{\theta_s}$ from the five methods (μK^2).

ℓ	D_ℓ^{TE} [μK^2]	CLASS	SPT num.	SPT anal.	lcdm v1 num.	lcdm v1 anal.
			$\partial D_\ell^{TE}/\partial\omega_m\|_{\theta_s}$ [μK^2]			
2	2.619	2.951	3.469	3.469	3.171	3.175
10	8.435×10^{-1}	2.652	2.568	2.568	2.453	2.453
30	1.891	3.012	2.866	2.866	3.024	3.023
100	-2.317×10^{1}	1.025×10^{2}	1.021×10^{2}	1.021×10^{2}	1.019×10^{2}	1.020×10^{2}
200	-1.537×10^{1}	-1.611×10^{2}	-1.611×10^{2}	-1.611×10^{2}	-1.613×10^{2}	-1.616×10^{2}
500	-5.863×10^{1}	3.718×10^{2}	3.722×10^{2}	3.722×10^{2}	3.729×10^{2}	3.725×10^{2}
800	-8.893×10^{1}	3.166×10^{2}	3.186×10^{2}	3.186×10^{2}	3.195×10^{2}	3.183×10^{2}
1000	-2.427×10^{1}	5.149×10^{1}	4.882×10^{1}	4.881×10^{1}	4.923×10^{1}	5.049×10^{1}
1500	-4.459×10^{-1}	-7.132	-7.161	-7.156	-6.820	-7.356
2000	-1.804×10^{1}	3.044×10^{1}	3.106×10^{1}	3.107×10^{1}	3.148×10^{1}	3.146×10^{1}
2500	-2.935	-1.396×10^{1}	-1.408×10^{1}	-1.408×10^{1}	-1.387×10^{1}	-1.387×10^{1}

Table 8: EE: D_ℓ and $\partial D_\ell/\partial\omega_m|_{\theta_s}$ from the five methods (μK^2).

ℓ	D_ℓ^{EE} [μK^2]	CLASS	SPT num.	SPT anal.	lcdm v1 num.	lcdm v1 anal.
			$\partial D_\ell^{EE}/\partial\omega_m\|_{\theta_s}$ [μK^2]			
2	3.079×10^{-2}	-2.057×10^{-1}	-2.041×10^{-1}	-2.041×10^{-1}	-2.031×10^{-1}	-2.031×10^{-1}
10	3.052×10^{-3}	2.132×10^{-2}	2.123×10^{-2}	2.123×10^{-2}	2.108×10^{-2}	2.109×10^{-2}
30	2.155×10^{-2}	-4.375×10^{-2}	-4.323×10^{-2}	-4.323×10^{-2}	-4.234×10^{-2}	-4.241×10^{-2}
100	7.769×10^{-1}	-8.172×10^{-1}	-7.977×10^{-1}	-7.977×10^{-1}	-8.016×10^{-1}	-8.030×10^{-1}
200	6.896×10^{-1}	-1.002	-9.883×10^{-1}	-9.883×10^{-1}	-9.902×10^{-1}	-9.900×10^{-1}
500	8.315	-3.937×10^{1}	-3.942×10^{1}	-3.942×10^{1}	-3.930×10^{1}	-3.925×10^{1}
800	1.559×10^{1}	-4.662×10^{1}	-4.684×10^{1}	-4.684×10^{1}	-4.682×10^{1}	-4.668×10^{1}
1000	4.213×10^{1}	-1.064×10^{2}	-1.069×10^{2}	-1.069×10^{2}	-1.065×10^{2}	-1.064×10^{2}
1500	1.272×10^{1}	3.386	3.534	3.535	3.399	3.293
2000	8.634	-1.190×10^{1}	-1.245×10^{1}	-1.245×10^{1}	-1.249×10^{1}	-1.239×10^{1}
2500	2.896	1.370	1.217	1.217	1.151	1.143

B Files

`debug/dcl_domegam_spt_vs_class/`: `compute_derivs.py` (CLASS + SPT, ∼35 min on 2 cores), `compute_v1.py` (lcdm v1 through the pipeline's `classy.py`), `theta_check.py`, `make_report.py` (figures and tables), `spt_emu.py` (numpy SPT networks with exact Jacobian), `cpstub/` (numpy stand-in for `cosmopower`, only needed without TensorFlow), `*.npz` (all derivatives).